PYTHON ACTIVITY-BASED LEARNING

README • Project Documentation • Academic Year 2026–2027

Project Overview

This repository contains the Python Programming Activity-Based Learning (ABL) assignment for the academic year **2026–2027**. The consolidated report presents 10 selected case studies. Each case study includes a problem statement, objectives, assumptions, algorithm, flowchart, pseudocode, Python source code, test cases, sample-output screenshot, and conclusion, following the assignment circular.

Project Objectives

- Understand practical problems using Python.
- Develop logical and computational thinking.
- Represent solutions using algorithms and flowcharts.
- Implement and test Python programs.
- Analyse program outputs and draw conclusions.

Selected Case Studies

The consolidated report contains the following selected case studies and titles:

1. Case Study 2 — Progressive Electricity Billing
2. Case Study 12 — Attendance Recovery Computation
3. Case Study 22 — Laboratory Chemical Inventory Validation
4. Case Study 34 — Student Feedback Sentiment Scoring
5. Case Study 42 — Student Marks Matrix Analysis
6. Case Study 51 — Bus Passenger Occupancy Simulation
7. Case Study 64 — Tuple-Based Course Registration Validation
8. Case Study 67 — Retail Order Dispatch and Return Analysis
9. Case Study 86 — Multi-Stage ATM Transaction Audit
10. Case Study 92 — Multi-Event Sports Scheduling and Scoring

Project File

`Python_ABL_A_Balaji_26001374_Case_Studies_Clear_Code_Black.pdf` — consolidated ABL report containing the 10 selected case studies, Python programs, test cases, sample-output screenshots, and flowcharts.

If the PDF is renamed in the repository, update the filename above so that it exactly matches the uploaded file.

Tools Used

- **Python** — program implementation and testing
- **GitHub** — repository hosting and submission
- **Markdown** — project documentation

Conclusion

This ABL project supports the development of Python programming skills, logical problem-solving, algorithm design, flowchart interpretation, and program testing through practical case studies.

Student Details

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Academic Year	2026–2027